

Filipe Pinarello Scalcon, Ph.D.

Research Associate

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SUMMARY

- Ph.D. in Electrical Engineering with a focus on control applied to power electronics for applications in industry, electric traction and renewable energy conversion
- Over 35 applied research publications, including 9 top-tier IEEE journal papers and 23 IEEE international conference proceedings (h-index 11)
- Over 10 years of engineering experience, including industry experience, academic research in world's leading programs and direct involvement in 4 industry-academia projects
- Over 3 years of experience at the post-doctoral level
- Strong background on the fields of control, power electronics, electric machines and optimization with advanced proficiency in the design and construction of experimental hardware
- Knowledge and training in RTDS hardware-in-the-loop systems as well as experience with battery testing hardware systems
- Extensive experience leading, supervising and mentoring undergraduate and graduate students in research projects
- Excellent communication, teamwork, and interpersonal skills as evidenced by collaborative research, and effective involvement in interdisciplinary projects while interacting with internal and external stakeholders
- Over three years of teaching experience in both college and undergraduate courses, delivering courses at the Federal University of Santa Maria and the University of Calgary
- I am eligible for registration as a Professional Engineer with APEGA
- I am a permanent resident of Canada

EDUCATION

Mar. 2019 – Sept. 2021	Federal University of Santa Maria – Santa Maria, RS, Brazil Ph.D. in Electrical Engineering <i>GPA: 3.96/4.0</i> <i>Thesis: "Performance Improvement of Switched Reluctance Motors Through Advanced Control Strategies"</i> <i>Advisor: Prof. Hilton Abílio Gründling</i>
Mar. 2017 – Feb. 2019	Federal University of Santa Maria – Santa Maria, RS, Brazil M.S. in Electrical Engineering <i>GPA: 3.98/4.0</i> <i>Thesis: "Contributions to the Driving and Control of the Switched Reluctance and Synchronous Reluctance Motors"</i> <i>Advisor: Prof. Hilton Abílio Gründling</i>

Aug. 2017 – Jan. 2019	Federal University of Santa Maria – Santa Maria, RS, Brazil Special Graduation Program for Professor Training for Professional Education
Mar. 2012 – Feb. 2017	Federal University of Santa Maria – Santa Maria, RS, Brazil B.S. in Electrical Engineering <i>Cum. Average:</i> 8.81/10 <i>Advisor:</i> Prof. Rodrigo Padilha Vieira <i>Academic honours:</i> Awarded for student with the highest cumulative average.

RESEARCH EXPERIENCE

Feb. 2024 – Present	Department of Electrical and Software Engineering, University of Calgary – Calgary, AB, Canada
Nov. 2025 – Present	Research Associate • Led technical direction of the microgrid research project, coordinating efforts across electrical and chemical engineering groups • Assumed full ownership of laboratory operations, including safety protocols, equipment maintenance, and onboarding of new team members • Supervised 2 PhD and 2 MSc students, providing research guidance and performance oversight • Collaborated directly with the principal investigator to define strategic priorities and long-term research direction • Expanded publication output and contributed to grant writing and project reporting
Feb. 2024 – Oct. 2025	Postdoctoral Associate • Contributed to the construction and optimization of a campus microgrid facility, incorporating solar panels, microinverters, a Chroma battery system, a propane generator, and a flow battery • Assisted in coordinating day-to-day laboratory operations, including safety compliance, personnel training, and system maintenance • Mentored and collaborated with 2 PhD and 2 MSc students on research related to electric machine control and 3-phase grid-tie systems • Supported the principal investigator in shaping the research roadmap • Authored and published research findings in peer-reviewed journals
Jan. 2022 – Jan. 2024	McMaster Automotive Resource Centre, McMaster University – Hamilton, ON, Canada Postdoctoral Fellow • Worked in three industry sponsored projects in teams of upwards of 15 people and in close proximity of the company engineers • Explored automatic code generation for TI DSPs using Simulink and transferred functionalities from legacy C code, later validated in an experimental setup

	• Implemented different high performance control techniques for SRMs, such as an initial position estimation algorithm for the sensorless operation, different current control strategies for high speed SRMs and techniques for RMS capacitor current reduction • Contributed to the development of an IoT edge device for machine vision and image recognition using an NVIDIA Jetson platform • Elaborated several technical reports, journal and conference papers • Assisted the ECE department in activities such as participating in the interview of candidates for faculty positions
Mar. 2012 – Sept. 2021	Federal University of Santa Maria – Santa Maria, RS, Brazil
Mar. 2019 – Sept. 2021	Graduate Research Assistant (Doctorate) • Proposed new methods for the advanced control of switched reluctance motors • Collaborated with external researchers from McMaster University • Published 8 high-impact journal papers during a 2.5-year period • Worked closely with a group of approximately 10 graduate students, acting as a mentor to some of them
Mar. 2017 – Feb. 2019	Graduate Research Assistant (Masters) • Initiated the research on switched reluctance and synchronous reluctance motors in the institution • Quoted, designed and built an experimental setup for synchronous reluctance machines • Led a project with a budget of approximately 15k USD
Mar. 2012 – Feb. 2017	Research Assistant (Undergraduate) • Assisted graduate students with the assembly of experimental setups • Surveyed control strategies for induction and permanent magnet synchronous motors • Researched the control of six-phase BLDC motors

INDUSTRY EXPERIENCE

Aug. 2016 – Jan. 2017	John Deere – Horizontina, RS, Brazil Product Verification and Validation (PV&V) Intern – Full time • Carried out the instrumentation of combines and planters using a plethora of sensors • Performed data acquisition and post processing analysis for engineers • Implemented a remote position control circuit for a hydraulic pump, allowing safe operation of the equipment during destructive tests • Tested parts from alternative suppliers seeking cost reduction • Awarded with continuous improvement goals achievement as a member of the PV&V department
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TEACHING EXPERIENCE

Feb. 2024 – April 2024 Jan. 2025 – April 2025	University of Calgary – Calgary, AB, Canada Sessional Instructor – Part time <u>ENEL 582 – Modelling and Control of Electric Machines and Drives (Winter 2025)</u> • Prepared and deliver lectures and tutorials, as well as administer and grade assignments and exams • Prepared and delivered laboratories, also training teaching assistants and technicians for experiments <u>ENEL 585 – Introduction to Power Electronics (Winter 2024)</u> • Prepared and delivered lectures, as well as administered and graded assignments and exams • Prepared and delivered laboratories, also being responsible for grading reports
Mar. 2019 – Dec. 2019 Aug. 2017 – Dec. 2018	Federal University of Santa Maria – Santa Maria, RS, Brazil Teaching Assistant – Part time • Taught classes on the courses of electric machines, laboratory of electric machines and electrical measurements • Assisted in preparing class materials and grading assignments • Helped with experimental classes, demonstrating significant hands-on experience
Aug. 2018 – Dec. 2018	Industrial Technical College of Santa Maria – Santa Maria, RS, Brazil Teaching Intern – Part time • Taught the course of electrical measurements as part of the mandatory internship program for the Special Graduation Program for Professor Training for Professional Education • Prepared class materials, as well as elaborated and graded assignments and tests • Instructed experimental classes focused on handling and performing electrical measurements with digital and analog equipment

COURSES AND CERTIFICATIONS

- RTDS Simulator Introductory Training Course by RTDS. Earned March 14, 2025.
- Deep Learning Specialization by DeepLearning.AI on Coursera. Earned October 17, 2023.
- Machine Learning Specialization by Stanford University & DeepLearning.AI on Coursera. Earned July 7, 2023

SKILLS

- Computer skills: MATLAB, Simulink, PSIM, PLECS, RTDS, RSCAD FX, C/C++, Python, TensorFlow, Microsoft Office, CorelDraw, Latex.
- Languages: Portuguese (Native), English (Fluent).

AWARDS AND HONORS

- IEEE Transactions on Transportation Electrification Outstanding Reviewer Recognition (2025)
- IEEE IAS CMD Outstanding Chapter Award (2020)
- IEEE IAS Student Travel Grant, 12th Seminar on Power Electronics and Control (SEPOC 2019)
- Academic honours, Federal University of Santa Maria and Chamber of Commerce, Industry and Services of Santa Maria (2017)

JOURNAL PUBLICATIONS

- J1. G. X. Prestes, **F. Pinarello Scalcon**, C. J. V. Filho, G. F. Fang and R. P. Vieira, "Optimized Discrete-Time LQI Current Control for Indirect Torque Regulation of Switched Reluctance Motors," in *IEEE Transactions on Power Electronics*, vol. 41, no. 6, pp. 9133-9145, June 2026, doi: 10.1109/TPEL.2026.3651188.
- J2. C. J. V. Filho, G. Fang, **F. Pinarello Scalcon**, A. M. Knight and R. P. Vieira, "Multisegment Adaptive Current Control of Switched Reluctance Motors," in *IEEE Transactions on Transportation Electrification*, vol. 12, no. 1, pp. 1091-1101, Feb. 2026, doi: 10.1109/TTE.2025.3625491.
- J3. **F. Pinarello Scalcon**, G. X. Prestes, G. Fang, C. J. V. Filho, H. A. Gründling, R. P. Vieira, and A. M. Knight, "A Fast Firing Angle Optimization Approach for Current-Controlled Switched Reluctance Generators in Wind Power Applications," in *Eletrônica de Potência*, vol. 30, p. e202550, Aug. 2025, doi: 10.18618/REP.e202550.
- J4. G. X. Prestes, **F. Pinarello Scalcon**, B. Nahid-Mobarakeh, and R. P. Vieira, "Enhancing SRM Performance through Multi-Platform Optimization and FPGA-Based Real-Time Simulation", in *Eletrônica de Potência*, vol. 29, p. e202442, Oct. 2024, doi: 10.18618/REP.e202442.
- J5. **F. Pinarello Scalcon**, G. Fang, C. J. V. Filho, S. Dhale and B. Nahid-Mobarakeh, "Optimization of Analytical TSFs for DC-Link Current Reduction in Switched Reluctance Motors," in *IEEE Open Journal of Industry Applications*, vol. 5, pp. 356-368, 2024, doi: 10.1109/OJIA.2024.3441308.
- J6. C. J. V. Filho, G. Fang, **F. Pinarello Scalcon**, R. P. Vieira and B. Nahid-Mobarakeh, "Study on QEMF Model and Adaptive Full-Order Observer Design for Universal Sensorless Control of IPMSMs," in *IEEE Transactions on Transportation Electrification*, vol. 10, no. 1, pp. 1335-1346, March 2024, doi: 10.1109/TTE.2023.3276953.
- J7. G. Fang, **F. Pinarello Scalcon**, C. J. V. Filho, D. Xiao, B. Nahid-Mobarakeh and A. Emadi, "A Unified Wide-Speed Range Sensorless Control Method for Switched Reluctance Machines Based on Unsaturated Reluctance," in *IEEE Transactions on Industrial Electronics*, vol. 70, no. 10, pp. 9903-9913, Oct. 2023, doi: 10.1109/TIE.2022.3224123.
- J8. **F. Pinarello Scalcon**, G. Fang, C. J. V. Filho, H. A. Gründling, R. P. Vieira and B. Nahid-Mobarakeh, "A Review on Switched Reluctance Generators in Wind Power Applications: Fundamentals, Control and Future Trends," in *IEEE Access*, vol. 10, pp. 69412-69427, 2022, doi: 10.1109/ACCESS.2022.3187048.
- J9. **F. Pinarello Scalcon**, R. P. Vieira and H. A. Gründling, "PSO-Based Fast Mechanical Parameters Estimation of Switched Reluctance Motor Drives," in *Journal of Control, Automation and Electrical Systems*, vol. 33, pp. 1286-1293, 2022, doi: 10.1007/s40313-022-00896-0.
- J10. T. S. Gabbi, M. B. de Araujo, L. R. Rocha, **F. Pinarello Scalcon**, H. A. Gründling and R. P. Vieira, "Discrete-time sliding mode controller based on backstepping disturbance compensation for robust current control of PMSM drives," in *ISA Transactions*, vol. 128, part A, pp. 581-592, 2022, doi: 10.1016/j.isatra.2021.10.032.

- J11. **F. Pinarello Scalcon**, G. Fang, R. P. Vieira, H. A. Gründling and A. Emadi, "Discrete-Time Super-Twisting Sliding Mode Current Controller with Fixed Switching Frequency for Switched Reluctance Motors," in IEEE Transactions on Power Electronics, vol. 37, no. 3, pp. 3321-3333, March 2022, doi: 10.1109/TPEL.2021.3116096.
- J12. G. Fang, **F. Pinarello Scalcon**, D. Xiao, R. P. Vieira, H. A. Gründling and A. Emadi, "Advanced Control of Switched Reluctance Motors (SRMs): A Review on Current Regulation, Torque Control and Vibration Suppression," in IEEE Open Journal of the Industrial Electronics Society, vol. 2, pp. 280-301, 2021, doi: 10.1109/OJIES.2021.3076807.
- J13. **F. Pinarello Scalcon**, C. R. D. Osório, G. G. Koch, T. S. Gabbi, R. P. Vieira, H. A. Gründling, R. C. L. F. Oliveira and V. F. Montagner, "Robust Control of Synchronous Reluctance Motors by Means of Linear Matrix Inequalities," in IEEE Transactions on Energy Conversion, vol. 36, no. 2, pp. 779-788, June 2021, doi: 10.1109/TEC.2020.3028568.
- J14. G. G. Koch, C. R. D. Osório, **F. Pinarello Scalcon**, T. S. Gabbi, R. C. L. F. Oliveira, and V. F. Montagner, "Robust Control by Means of LMIs Applied to Permanent Magnet Synchronous Motors with Uncertain Parameters," in Eletrônica de Potência (Open Journal of Power Electronics), vol. 26, no. 1, pp. 115-124, 2021, doi: 10.18618/REP.2021.1.0066.
- J15. **F. Pinarello Scalcon**, T. S. Gabbi, R. P. Vieira and H. A. Gründling, "Performance Improvement of Switched Reluctance Motors via Particle Swarm Algorithm," in Eletrônica de Potência (Open Journal of Power Electronics), vol. 25, no. 4, pp. 492-502, 2020, doi: 10.18618/REP.2020.4.0038.
- J16. C. R. D. Osório, **F. Pinarello Scalcon**, G. G. Koch, V. F. Montagner, R. P. Vieira and H. A. Gründling, "Robust Control Applied to Switched Reluctance Generators Connected to the Grid," in Eletrônica de Potência (Open Journal of Power Electronics), vol. 25, no. 3, pp. 272-282, 2020, doi: 10.18618/REP.2020.3.0015.

JOURNAL PUBLICATIONS IN PREPARATION

- 1. G. X. Prestes, W. K. Moreira, **F. Pinarello Scalcon**, A. B. Júnior, A. M. Knight, C. Rech and R. P. Vieira, "PPO-Based Deep Reinforcement Learning Strategy for Current Control of Switched Reluctance Motors," in IEEE Transactions on Industrial Applications (In preparation).
- 2. S. A. Seif Kashani, **F. Pinarello Scalcon** and A. M. Knight, "Robust Control of Grid-Tied and Islanded Virtual Power Plants over 5G Network," in IEEE Transactions on Industrial Electronics (In preparation).

INTERNATIONAL CONFERENCE PROCEEDINGS

- P1. **F. Pinarello Scalcon**, G. X. Prestes, R. P. Vieira and A. M. Knight, "Optimization-Based Capacitor Current Reduction in Switched Reluctance Generators Operating under Single-Pulse Control," *2025 17th Seminar on Power Electronics and Control (SEPOC)*, Balneário Camboriú, Brazil, 2025, pp. 1-7, doi: 10.1109/SEPOC67005.2025.11297596.
- P2. G. X. Prestes, P. H. d. S. e. Silva, M. S. Lorenzo, **F. Pinarello Scalcon**, A. M. Knight and R. P. Vieira, "Nested Optimization-Based Current Shaping for Torque Ripple and Losses Mitigation in SRMs," *2025 17th Seminar on Power Electronics and Control (SEPOC)*, Balneário Camboriú, Brazil, 2025, pp. 01-08, doi: 10.1109/SEPOC67005.2025.11297685.
- P3. **F. Pinarello Scalcon**, G. X. Prestes, R. P. Vieira and A. M. Knight, "An Approach for RMS Capacitor Current Reduction in Current-Controlled Switched Reluctance Generators," *2025*

- IEEE Energy Conversion Conference Congress and Exposition (ECCE)*, Philadelphia, PA, USA, 2025, pp. 1-7, doi: 10.1109/ECCE58356.2025.11260047.
- P4. S. S. Kashani, **F. Pinarello Scalcon** and A. M. Knight, "Experimental Validation of Hybrid Local and Remote Supervisory Control of Virtual Power Plants Over 5G Cellular Networks," *2025 IEEE Energy Conversion Conference Congress and Exposition (ECCE)*, Philadelphia, PA, USA, 2025, pp. 1-6, doi: 10.1109/ECCE58356.2025.11260245.
- P5. A. Singh, **F. Pinarello Scalcon** and A. M. Knight, "Optimized Discrete-Time Super-Twisting Sliding Mode Control for Brushless Doubly Fed Reluctance Machines," *2025 IEEE Energy Conversion Conference Congress and Exposition (ECCE)*, Philadelphia, PA, USA, 2025, pp. 1-7, doi: 10.1109/ECCE58356.2025.11260340.
- P6. S. S. Kashani, **F. Pinarello Scalcon** and A. M. Knight, "Control of Grid Connected Virtual Power Plants Over 5G Cellular Networks by a Hybrid Local and Supervisory Approach," *2025 IEEE Industry Applications Society Annual Meeting (IAS)*, Taipei, Taiwan, 2025, pp. 1-6, doi: 10.1109/IAS62731.2025.11061675.
- P7. A. Singh, **F. Pinarello Scalcon** and A. M. Knight, "Challenges in the Control of Brushless Doubly Fed Reluctance Machines," *2025 IEEE Industry Applications Society Annual Meeting (IAS)*, Taipei, Taiwan, 2025, pp. 1-7, doi: 10.1109/IAS62731.2025.11061502.
- P8. G. X. Prestes, W. K. Moreira, **F. Pinarello Scalcon**, C. Rech, A. M. Knight and R. P. Vieira, "Reinforcement Learning-Based Current Controller for Switched Reluctance Motor Drives," *2025 IEEE International Electric Machines & Drives Conference (IEMDC)*, Houston, TX, USA, 2025, pp. 71-76, doi: 10.1109/IEMDC60492.2025.11061049.
- P9. **F. Pinarello Scalcon**, I. G. Zurbriggen, A. M. Knight, "Super-Twisting Sliding Mode Output Voltage Controller for Switched Reluctance Generators in Wind Power Applications," *IECON 2024 – 50th Annual Conference of the IEEE Industrial Electronics Society*, Chicago, IL, USA, 2024, pp. 1-6, doi: 10.1109/IECON55916.2024.10905393.
- P10. **F. Pinarello Scalcon**, R. Tahal, M. Ahrabi, Y. Huangfu, R. Ahmed, B. Nahid-Mobarakeh, S. Shirani, C. Vidal and A. Emadi, "AI-Powered Video Monitoring: Assessing the NVIDIA Jetson Orin Devices for Edge Computing Applications," *2024 IEEE Transportation Electrification Conference and Expo (ITEC)*, Chicago, IL, USA, 2024, pp. 1-6, doi: 10.1109/ITEC60657.2024.10598994.
- P11. **F. Pinarello Scalcon**, G. Fang, C. J. V. Filho, H. A. Gründling, R. P. Vieira and B. Nahid-Mobarakeh, "An Asymmetric Five-Level Flying Capacitor Converter with PWM Current Control for Switched Reluctance Motors," *2023 IEEE 8th Southern Power Electronics Conference and 17th Brazilian Power Electronics Conference (SPEC/COBEP)*, Florianopolis, Brazil, 2023, pp. 1-6, doi: 10.1109/SPEC56436.2023.10408124.
- P12. G. X. Prestes, A. B. Júnior, **F. Pinarello Scalcon**, L. R. Rocha, W. B. Da Silveira, H. A. Grundling, "Discrete-Time Higher-Order Sliding Mode Current Controller for Switched Reluctance Motor Drives," *2023 IEEE 8th Southern Power Electronics Conference and 17th Brazilian Power Electronics Conference (SPEC/COBEP)*, Florianopolis, Brazil, 2023, pp. 1-8, doi: 10.1109/SPEC56436.2023.10407360.
- P13. **F. Pinarello Scalcon**, G. Fang, C. J. V. Filho, H. A. Gründling, R. P. Vieira and B. Nahid-Mobarakeh, "A PWM Fixed-Gain Super-Twisting Sliding Mode Current Controller for Switched Reluctance Motors," *IECON 2022 – 48th Annual Conference of the IEEE Industrial Electronics Society*, Brussels, Belgium, 2022, pp. 1-6, doi: 10.1109/IECON49645.2022.9968860.
- P14. G. Fang, **F. Pinarello Scalcon**, D. Xiao, B. Nahid-Mobarakeh and A. Emadi, "Simultaneous Radial Force and Torque Control for Switched Reluctance Motors Based on Optimized Quadratic Sharing Function Method," *IECON 2022 – 48th Annual Conference of the IEEE Industrial*

- Electronics Society, Brussels, Belgium, 2022, pp. 1-6, doi: 10.1109/IECON49645.2022.9968639.
- P15. C. J. V. Filho, **F. Pinarello Scalcon**, R. P. Vieira and B. Nahid-Mobarakeh, "A New β -axis Based High-Frequency Signal Injection Model for Low-Speed Sensorless IPMSM Drives," IECON 2022 – 48th Annual Conference of the IEEE Industrial Electronics Society, Brussels, Belgium, 2022, pp. 1-6, doi: 10.1109/IECON49645.2022.9968828.
- P16. I. Rasoanarivo, K. Tehrani, **F. Pinarello Scalcon** and B. Nahid-Mobarakeh, "A Dual Multilevel Adaptive Converter for Microgrid Applications," IECON 2022 – 48th Annual Conference of the IEEE Industrial Electronics Society, Brussels, Belgium, 2022, pp. 1-6, doi: 10.1109/IECON49645.2022.9968905.
- P17. **F. Pinarello Scalcon**, R. P. Vieira and H. A. Gründling, "PSO-Based Performance Optimization Procedure for Current-Controlled Switched Reluctance Generators in Wind Power Applications," IECON 2021 – 47th Annual Conference of the IEEE Industrial Electronics Society, Toronto, ON, Canada, 2021, pp. 1-6, doi: 10.1109/IECON48115.2021.9589390.
- P18. C. R. D. Osório, **F. Pinarello Scalcon**, R. P. Vieira, V. F. Montagner and H. A. Gründling, "Robust Control of Switched Reluctance Generator In Connection With a Grid-Tied Inverter," 2019 IEEE 15th Brazilian Power Electronics Conference and 5th IEEE Southern Power Electronics Conference (COBEP/SPEC), Santos, Brazil, 2019, pp. 1-6, doi: 10.1109/COBEP/SPEC44138.2019.9065551.
- P19. T. Lazzari, **F. Pinarello Scalcon**, C. Volpato, T. Gabbi, M. Stefanelo and R. P. Vieira, "Sensorless Control of Nonsinusoidal Back-EMF PMSM Based on State Observer," 2019 IEEE 15th Brazilian Power Electronics Conference and 5th IEEE Southern Power Electronics Conference (COBEP/SPEC), Santos, Brazil, 2019, pp. 1-6, doi: 10.1109/COBEP/SPEC44138.2019.9065788.
- P20. **F. Pinarello Scalcon**, C. J. Volpato, T. Lazzari, T. S. Gabbi, R. P. Vieira and H. A. Gründling, "Sensorless Control of a SynRM Drive Based on a Luenberger Observer with an Extended EMF Model," IECON 2019 - 45th Annual Conference of the IEEE Industrial Electronics Society, Lisbon, Portugal, 2019, pp. 1333-1338, doi: 10.1109/IECON.2019.8927192.
- P21. **F. Pinarello Scalcon**, T. S. Gabbi, R. P. Vieira and H. A. Gründling, "Decoupled Vector Control Based on Disturbance Observer Applied to the Synchronous Reluctance Motor," 2019 21st European Conference on Power Electronics and Applications (EPE '19 ECCE Europe), Genova, Italy, 2019, pp. P.1-P.8, doi: 10.23919/EPE.2019.8915498.
- P22. **F. Pinarello Scalcon**, R. P. Vieira and H. A. Gründling, "Sliding Mode Speed Control Applied to the Switched Reluctance Motor," IECON 2018 - 44th Annual Conference of the IEEE Industrial Electronics Society, Washington, DC, USA, 2018, pp. 695-700, doi: 10.1109/IECON.2018.8591829.
- P23. C. J. Volpato Filho, **F. Pinarello Scalcon**, T. S. Gabbi and R. P. Vieira, "Adaptive observer for sensorless permanent magnet synchronous machines with online pole placement," 2017 Brazilian Power Electronics Conference (COBEP), Juiz de Fora, Brazil, 2017, pp. 1-6, doi: 10.1109/COBEP.2017.8257406.

SERVICE

SERVICE TO PROFESSION/VOLUNTEERING:

- Associate editor, *Eletrônica de Potência* (Open Journal of Power Electronics) (Oct 2024 – present)
- Session chair - Multiphase Electric Drives for Transportation Electrification, 50th Annual Conference of the IEEE Industrial Electronics Society (IECON 2024)

- Session chair - AI based Power Electronics and Electric Drives for Enhanced Performance, 50th Annual Conference of the IEEE Industrial Electronics Society (IECON 2024)
- Mentor, University of Calgary Solar Car Team (2024)
- Guest Editor - Special Issue "Advanced Control Methods and Emerging Motor Designs for Electric Propulsion Applications", Electronics by MDPI (2023)
- Track chair, 15th Seminar on Power Electronics and Control (SEPOC 2023)
- Track chair, 2021 IEEE International Power and Renewable Energy Conference (IPRECON 2021)
- Vice president, IEEE IAS UFSM Student Chapter (2019)
- Organizing committee, 11th Seminar on Power Electronics and Control (SEPOC 2018)
- Organizing committee, 10th Seminar on Power Electronics and Control (SEPOC 2017)
- Organizing committee, 9th Seminar on Power Electronics and Control (SEPOC 2015)

REVIEWER:

- IEEE Transactions on Power Electronics
- IEEE Transactions on Industrial Electronics
- IEEE Transactions on Transportation Electrification
- IEEE Transactions on Mechatronics
- IEEE Journal of Emerging and Selected Topics in Power Electronics
- IEEE Journal of Emerging and Selected Topics in Industrial Electronics
- IEEE Open Journal of the Industrial Electronics Society
- IEEE Open Journal of Power Electronics
- IEEE Access
- IET Power Electronics
- IET Electric Power Applications

PROFESSIONAL MEMBERSHIPS:

- IEEE Member, 2017 – Present
- IEEE Industry Application Society, Total years - 6
- IEEE Industrial Electronics Society, Total years - 4
- IEEE Power Electronics Society, Total years - 5

REFERENCES

Prof. Ali Emadi, Ph.D., Fellow IEEE, SAE, and NAI (Postdoc Supervisor)

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